

ICST INETRDISCIPLINARY CENTER FOR SMART TECHNOLOGIES
Center of CS Department's Software Engineering Track

SmartRail

Know where to board.

Problem: Passengers lack seat information before boarding.
They may crowd near one door or search inside the train.

SmartRail: Our IoT prototype monitors train-seat availability.
It guides passengers toward cars with more free seats.

- **Seat sensing:** ESP32 and IR sensors detect occupied seats.
- **Hybrid train:** 12 x 108 seats (1,296); two physical inputs.
- **Web guidance:** Car counts, seat maps and zone suggestions.

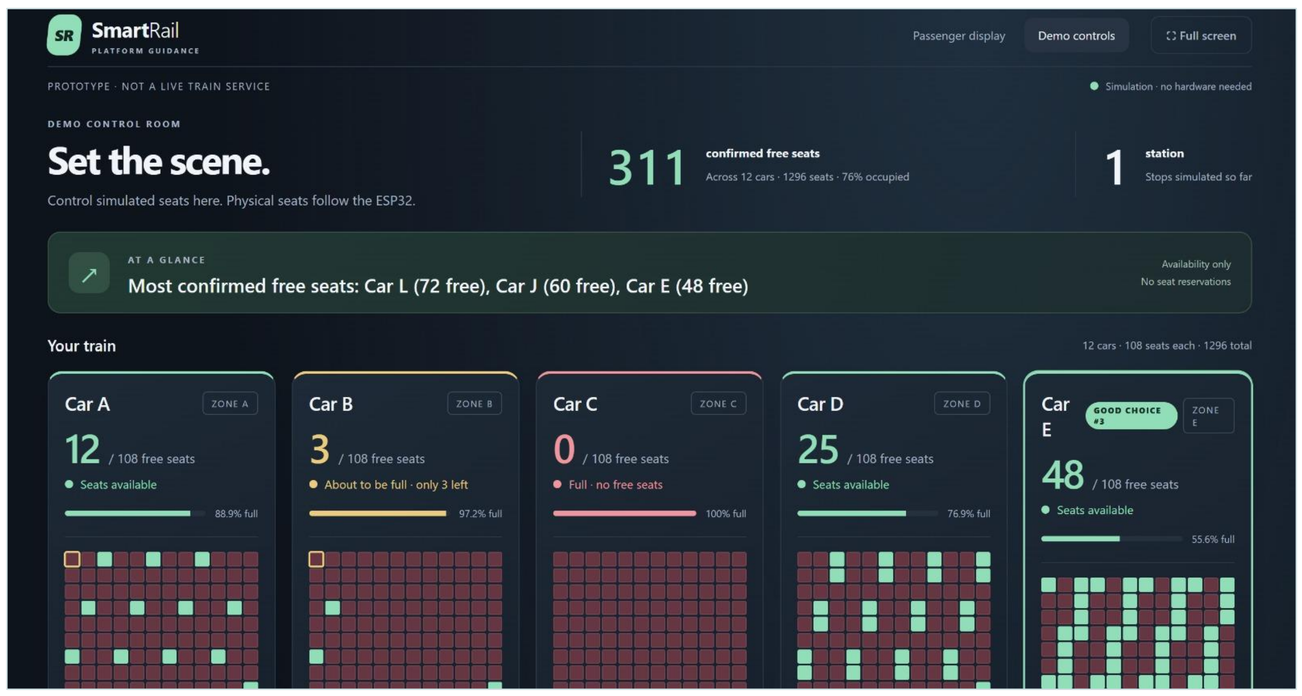
Advantage: Check before boarding, not after entering a car.

Demonstration: 2 physical seats + 1,294 simulated seats.

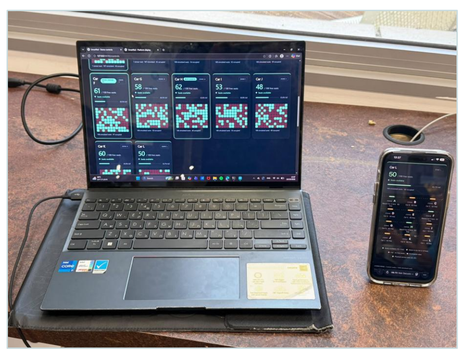
Conclusion: One display combines real and simulated seats.

METHOD / PROCESS — FROM SENSING TO GUIDANCE

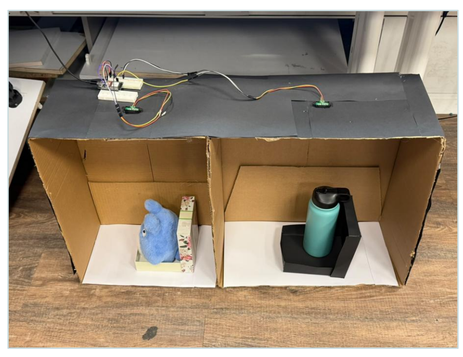
1. Calibrate At startup → 2. Detect Filtered IR → 3. Send USB state → 4. Merge Real + sim. → 5. Count Free / car → 6. Guide Show zones



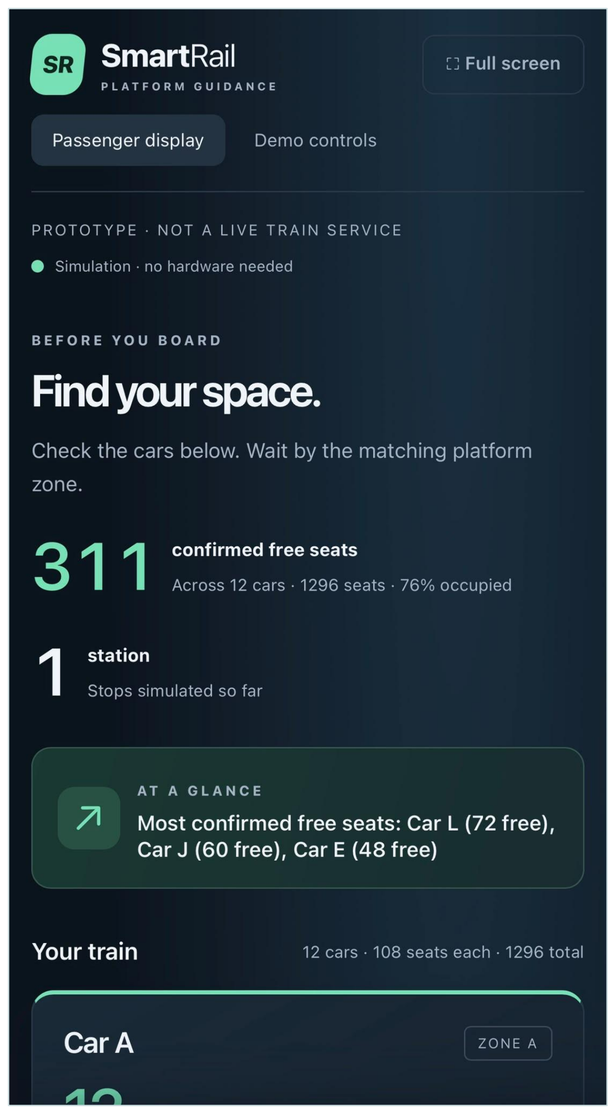
Desktop controls (simulation example)



Laptop + mobile display



Two sensor seat models



Passenger view (simulation)

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github.com/abedazem13/SmartRail

A Project in Internet of Things (IoT)

Computer Science Department
הטכניון מכון טכנולוגי לישראל